

Reglas de la derivación

Funciones

Funciones lineales

Funciones exponenciales

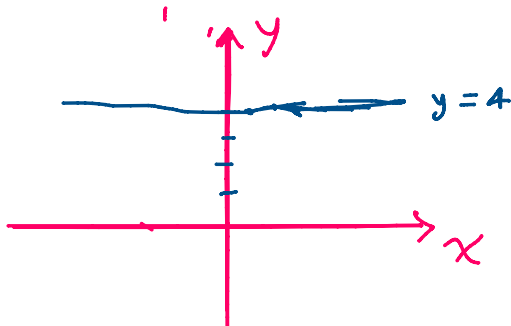
Funciones Potencia

Funciones racionales

⋮

Reglas de la derivación

1. Función constante.



$$f(x) = 4$$

$$f(x) = y = \underline{4}$$

derivada de una constante $\frac{d(c)}{dx} = 0$

2. Función Potencia

x^2, x^3, x^{1000}
~~~~~  
Potencias

$$\frac{d(cx^n)}{dx} = \underline{cnx^{n-1}}$$

$c = \text{constante}$

$n = \text{Potencia.}$

$$f(x) = y = \underline{6x^6} \quad f', \frac{dy}{dx}, \frac{df(x)}{dx}$$

$$y' = 6x^5 \quad \begin{matrix} a=1 \\ n=5 \end{matrix}$$

$$y' = 6x^5$$

$$f(x) = 2x^4$$

$$f'(x) = 8x^3 \quad \checkmark$$

$x^n$   $\nabla$  potencia

$$f'(x) = n x^{n-1}$$

Derivadas  $f(x) = x^2$

$$f'(x) = 2x^{2-1} = 2x^1 = 2x$$

$$f(x) = 5x^8$$

$$f'(x) = 40x^7$$

$$f(x) = x^1$$

$$f'(x) = 1x^{1-1} = x^0 = 1$$

Todo # elevado a la cero  
es igual a 1

$$\begin{cases} f(x) = x \\ f'(x) = 1 \end{cases}$$

$$a^{-n} = \frac{1}{a^n}$$

$$g(x) = \frac{1}{x^3}$$

$$g(x) = x^{-3}$$

$$g'(x) = -3x^{-3-1} = -3x^{-4}$$

$$g'(x) = \frac{-3}{x^4}$$

$$h(x) = \sqrt[2]{x^1} \rightarrow \text{¿La puedo expresar como potencia?}$$

$$h(x) = x^{1/2}$$

$$\sqrt[n]{a^n} = a^{n/m}$$

$$h'(x) = \frac{1}{2} x^{\frac{1}{2}-1} = \frac{1}{2} x^{-\frac{1}{2}}$$

$$\begin{aligned} f(x) &= \sqrt[4]{x} \\ f'(x) &= \frac{1}{4} x^{-3/4} \end{aligned}$$

$$\begin{aligned} f(x) &= x^{1/4} \\ f'(x) &= \frac{1}{4} x^{\frac{1}{4}-1} \end{aligned}$$

$$\frac{1}{4} \cancel{x^1} = \frac{1-1}{4} = \frac{-3}{4}$$

$$p(x) = 3x^2 + 8x + 1$$

$$P(x) = 3x^2 + 8x + 1$$

$$P'(x) = 6x + 8 + 0$$

$$P'(x) = 6x + 8$$

$$\frac{1}{\sqrt{x}} = \frac{1}{x^{1/2}} = x^{-1/2}$$

$$P(x) = 3x^5 + \frac{1}{\sqrt{x}} - \frac{1}{x} - 8$$

$$\frac{-1}{2} - 1 =$$

$$P'(x) = 15x^4 - \frac{1}{2}x^{-3/2} - (-x^{-2})$$

$$P'(x) = 15x^4 - \frac{1}{2x^{3/2}} + \frac{1}{x^2}$$

Formula general para derivar una potencia

Derivada de un polinomio  $\leadsto$  Derivada de cada uno de sus términos

Regla de la suma y Regla de la diferencia

$$(f+g)' = f' + g'$$

$$(f-g)' = f' - g'$$

comilla  
↓  
Derivada

$f(x)$

$\hookrightarrow$  Derivada

$\hookrightarrow f'(x)$

$$\bullet \frac{d f(x)}{d x}$$

$$\bullet \frac{d y}{d x}$$

$y'$

$$\boxed{y = f(x)}$$

$$\frac{d}{d x} c x^n = c n x^{n-1}$$

$$f(x) = 5x^3 - 4x^4 - x^2 - 18x + 10$$

$$f'(x) = 15x^2 - 16x^3 - 2x - 18$$

$$18x^1 \leadsto 1 \cdot 18x^{1-1} = 18x^0 = 18$$

## Regla del producto

Si  $g$  y  $h$  son derivables

↓ Función

$$f(x) = g(x)h(x)$$

↑

$$P(x) = (2x^2 - 3)(x^5 + 7x^2)$$

$$P'(x) = (4x)(x^5 + 7x^2) + (5x^4 + 14x)(2x^2 - 3)$$

$$f'(x) = g'(x)h(x) + h'(x)g(x)$$

$$f'(e) = 0$$

$$f(x) = x e^x$$

$e^x \rightarrow$  función exponencial natural

Euler  $\rightarrow 2,71828$

$$\left\{ \frac{d e^x}{d x} = e^x \right\}$$

$$f'(x) = g'(x)h(x) + h'(x)g(x)$$

$$f'(x) = 1(e^x) + e^x x$$

$$= e^x + x e^x$$

$$= e^x (1+x) \checkmark$$

## Regla del cociente

$$y = f(x) = \frac{g(x)}{h(x)} \quad f'(x) = \frac{g'(x)h(x) - h'(x)g(x)}{[h(x)]^2}$$

$$y' = \frac{g'h - h'g}{h^2}$$

$$y = \frac{x^2 + x - 2}{x^3 + 6}$$

$\rightarrow g$

$\rightarrow h$

$$\begin{cases} f(x) = x \\ f'(x) = 1 \end{cases}$$

$$y' = \frac{(2x+1)(x^3+6) - (3x^2)(x^2+x-2)}{(x^3+6)^2}$$

n. dentro de

$$v \quad \frac{\quad}{(x^3+6)^2}$$

Regla de la cadena  $\rightarrow$  Cuando tengo una fn dentro de otra función

$$y = H(U(x)) \quad \frac{dy}{dx} = H'(U(x)) \cdot U'(x)$$

$\downarrow$

•  $y = x^n \rightarrow y' = nx^{n-1}$

•  $p(x) = x^{n+1} + x^n + x^{n-1}$

$y = (3x+5)^3$   $\rightarrow$  Potencia 3

$y' = 3(3x+5)^{3-1} \cdot 3$

$y' = 9(3x+5)^2$

Cambio de términos

$u(x) = 3x+5 \rightarrow u'(x) = 3$

$h(x) = u^3 \rightarrow h'(x) = 3u^2$

$\frac{dy}{dx} = 3u^2 \cdot 3$

$\downarrow$   
9

$\frac{dy}{dx} = 9u^2$

$\hookrightarrow u?$

$\frac{dy}{dx} = 9(3x+5)^2$

$f(x) = (e^x + 2x)^2$

$f'(x) = h'(u(x)) \cdot u'(x)$

$f'(x) = 2u \cdot (e^x + 2)$

$u(x) = e^x + 2x$

$h(x) = u^2$

$u'(x) = e^x + 2$

$h'(u(x)) = 2u$

$$f'(x) = 2(e^x + 2x) \cdot (e^x + 2)$$

Derivada de un logaritmo

$$y = \text{Log}_a(h(x))$$

$$y' = \frac{1}{\ln a} \cdot \frac{h'(x)}{h(x)}$$

$$y = \text{Log}_5(x^3 + 2x)$$

$$y' = \frac{1}{\ln 5} \cdot \frac{(3x^2 + 2)}{x^3 + 2x}$$

$$h(x) = y$$

$$f(x) = y$$

$$g(x) = y$$

Derivada del Logaritmo natural

$$f(x) = \ln(h(x))$$

$$f'(x) = \frac{h'(x)}{h(x)}$$

$$f(x) = \ln(x^2)$$

$$\hookrightarrow h(x) = x^2$$

$$f'(x) = \frac{2x}{x^2} = \frac{2\cancel{x}}{\cancel{x} \cdot x} = \frac{2}{x}$$

$$h'(x) = 2x$$

$$f(x) = \ln(h(x))$$

$$f'(x) = \frac{h'(x)}{h(x)}$$

$$\frac{dy}{dx} = H'(u(x)) \cdot u'(x)$$

$$y' = \frac{g'h - h'g}{h^2}$$

$$f(x) = \ln\left(\frac{x+1}{\sqrt{x-2}}\right)$$

$$f(x) = \ln\left(\frac{x+1}{\sqrt{x-2}}\right)$$

$$f(x) = \ln(u)$$

$$f'(x) = \frac{u'}{u}$$

$$\ln(u)$$

$$u = \frac{x+1}{\sqrt{x-2}} \quad \left\} \frac{g}{h} \rightarrow \left[ h' = \sqrt{x-2} = (x-2)^{1/2} = \frac{1}{2} (x-2)^{1/2-1} = \frac{1}{2} (x-2)^{-1/2} = \frac{1}{2\sqrt{x-2}} \right]$$

$$u' = 1(\sqrt{x-2}) - \frac{1}{2\sqrt{x-2}} \cdot (x+1) = \frac{\sqrt{x-2} - \left(\frac{x+1}{2\sqrt{x-2}}\right)}{(x-2)^2}$$

$$f'(x) = \frac{\sqrt{x-2} - \left(\frac{x+1}{2\sqrt{x-2}}\right)}{x-2}$$

$$f'(x) = \frac{\sqrt{x-2} \left( \sqrt{x-2} - \left(\frac{x+1}{2\sqrt{x-2}}\right) \right)}{(x-2)(x+1)} = \frac{x-2 - \frac{x+1}{2}}{(x-2)(x+1)} = \frac{x-5}{2(x+1)(x-2)}$$